RAPS-5 V1

Redundant Avionics Power Supervisor

2S-6S LiPo A -> 5.200V -> LTC4412 + PMOS

PATH A



Αρχείο: Path_A.kicad_sch

2S-6S LiPo B -> 5.152V -> LTC4412 + PMOS

PATH B



Αρχείο: Path_B.kicad_sch

5V_BUS: nominal ~5.2V / 3A

OUTPUT / SUPERVISION



Αρχείο: Path_O.kicad_sch

A/B positive rails join only after the ideal-diode MOSFETs. Shared solid ground plane.

A favored nominally. 50mV offset does not guarantee priority across regulator tolerances.

Experimental UAV / robotics hardware. Bench qualification required before operational use.

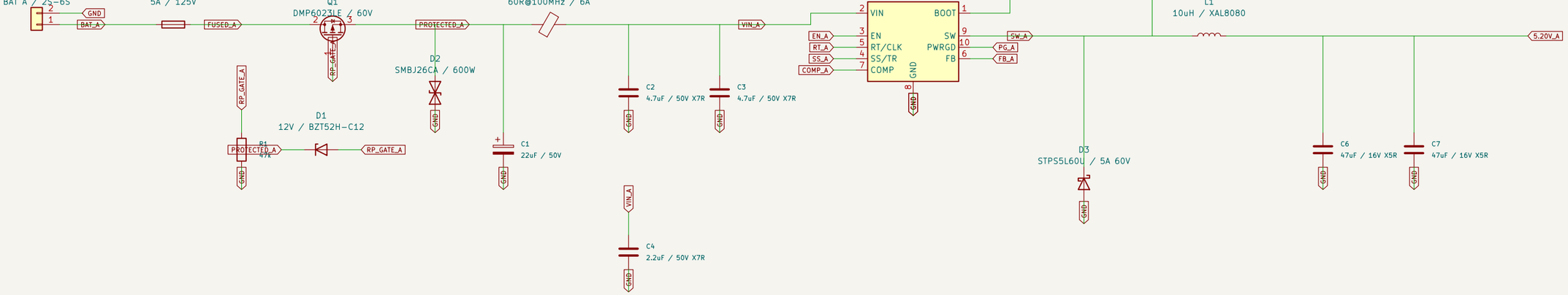
Experimental UAV / robotics module

Sheet: /
File: RAPS-5_V1.kicad_sch

Title: RAPS-5 V1 - Redundant Avionics Power Supervisor

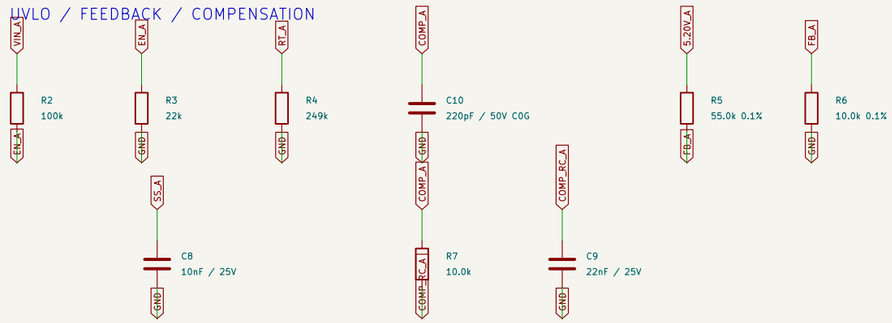
Size: A4	Date: 2026-10-01	**Rev: V1**
KiCad E.D.A. 10.0.5		Id: 1/4

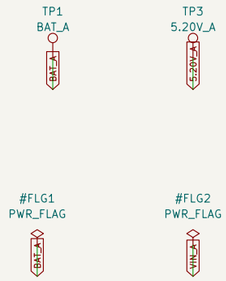
1. INPUT / FUSE / REVERSE POLARITY / TVS / EMI



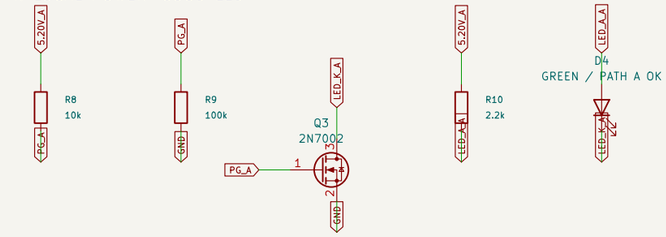
2. 400kHz BUCK – EACH PATH CAPABLE OF 3A

3. UVLO / FEEDBACK / COMPENSATION

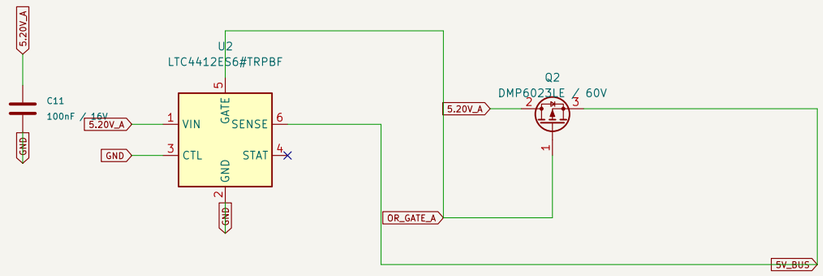




4. LOCAL POWER–GOOD LED



5. LTC4412 IDEAL DIODE – BLOCKS BUS BACKFEED



PMOS drain is upstream. Body diode points toward protected output / BUS.

PG LED means converter in regulation; it does not indicate current sharing.

RFB: VOUT = 0.8V x (1 + Rtop / 10k). Path A: 5.200V; Path B: 5.152V.

Common GND. Positive paths independent until ideal–diode OR–ing.

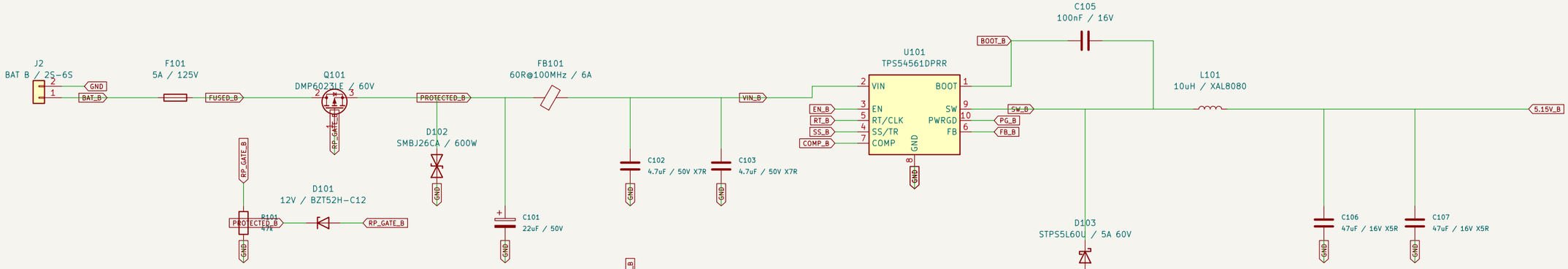
Experimental UAV / Robotics

Sheet: /PATH A/
File: Path_A.kicad_sch

Title: RAPS–5 V1 – PATH A

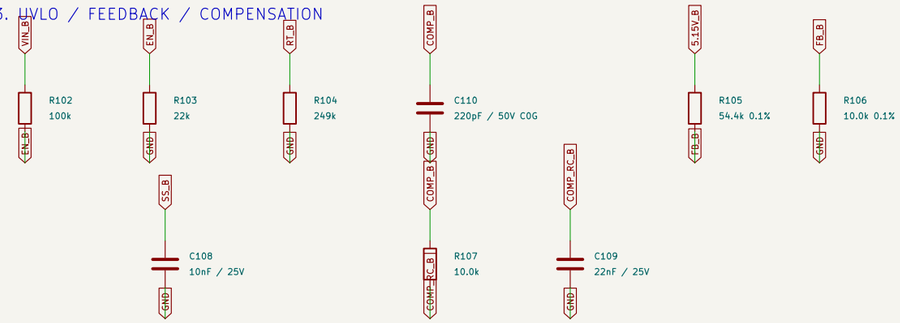
Size: A3	Date: 2026–10–01	**Rev: V1**
KiCad E.D.A. 10.0.5		Id: 2/4

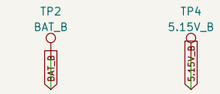
1. INPUT / FUSE / REVERSE POLARITY / TVS / EMI



2. 400kHz BUCK – EACH PATH CAPABLE OF 3A

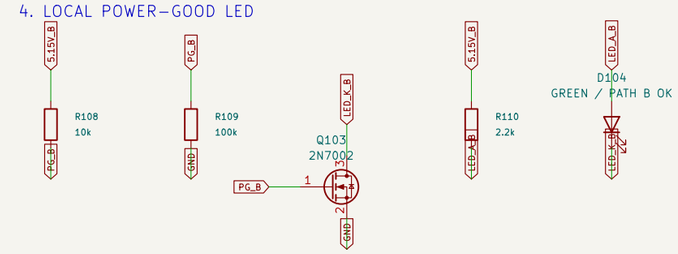
3. UVLO / FEEDBACK / COMPENSATION



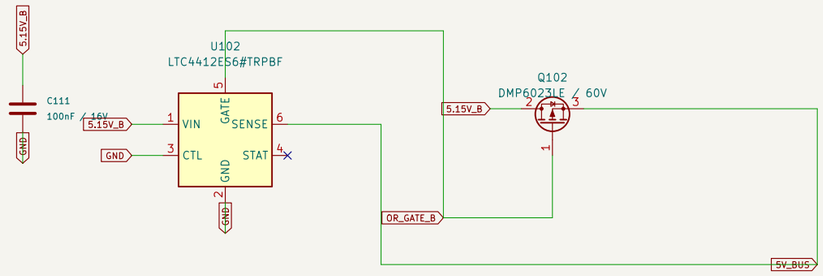




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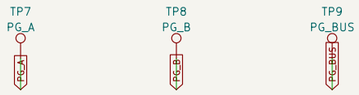
Experimental UAV / Robotics

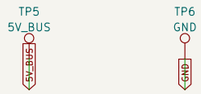
Sheet: /PATH B/
File: Path_B.kicad_sch

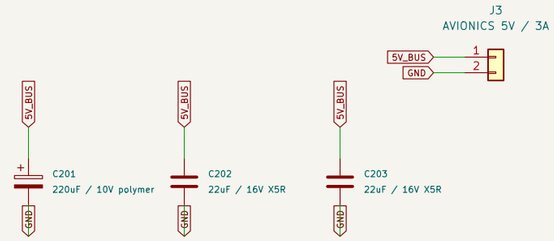
Title: RAPS–5 V1 – PATH B

Size: A3	Date: 2026–10–01	**Rev: V1**
KiCad E.D.A. 10.0.5		Id: 3/4

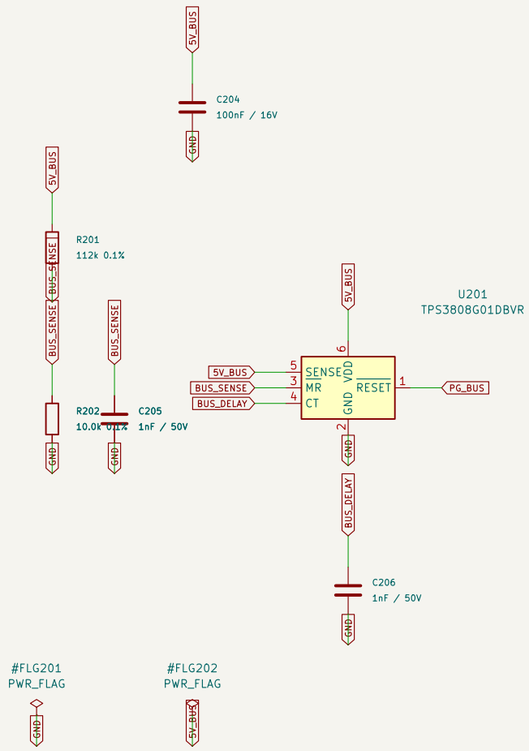
COMMON OUTPUT — 3A CONTINUOUS DESIGN TARGET



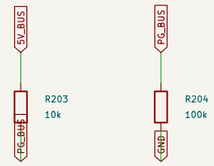


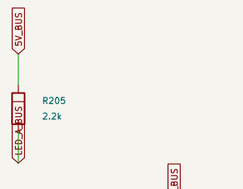


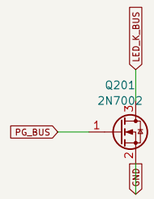
OUTPUT SUPERVISOR: 4.941V nominal threshold / ~6ms delay



OUTPUT OK: voltage qualified by TPS3808G01











All grounds common. No galvanic isolation; independent positive power channels.

Experimental prototype: validate thermal rise, stability, switchover and reverse current.

Common GND. Positive paths independent until ideal-diode OR-ing.

Experimental UAV / Robotics

Sheet: /OUTPUT / SUPERVISION/
File: Path_O.kicad_sch

Title: RAPS-5 V1 — OUTPUT / SUPERVISION

Size: A3	Date: 2026-10-01	**Rev: V1**
KiCad E.D.A. 10.0.5		Id: 4/4